

Lesson 6

Chapter 3: Classical demand theory — preferences, axioms, and utility

1 The classical approach

We build demand behavior from scratch, by imposing fundamental axioms on consumer preferences and seeing what these assumptions imply about behavior.

2 The preference relation

The primitive of the model is the preference relation $\succsim$. This is hard-wired into the consumer and captures everything about his taste or preference.

If $x, y \in X$ with $x \succsim y$, then x is w.p. (weakly preferred) to y . From $\succsim$:

- if $x \succsim y$ but not $y \succsim x$, then $x \succ y$;
- if $x \succsim y$ and $y \succsim x$, then the consumer is indifferent, $x \sim y$.

So we build $\succ$ and $\sim$ from $\succsim$; the latter is the primitive.

3 The basic axioms: completeness and transitivity

We start with axioms that imply that the consumer's preferences are relations in the basic sense.

- *Completeness*: for $x, y \in X$, either $x \succsim y$ or $y \succsim x$, or both.
- *Transitivity*: if $x \succsim y$ and $y \succsim z$, then $x \succsim z$.

This just tells us that the consumer is rational, but it does not tell us about what his tastes are like. The remaining axioms have more economic meaning.

4 Local non-satiation

Local non-satiation. For every $x \in X$ and any $\varepsilon > 0$ there exists $y \in X$ such that $\|x - y\| < \varepsilon$ and $y \succ x$.

So there is no bliss point: the consumer is never satiated, because he can always get a better bundle close by. Two important implications:

- Indifference curves are thin — kill the thick zone.
- The consumer, when he optimizes, spends his entire income: the budget constraint binds.

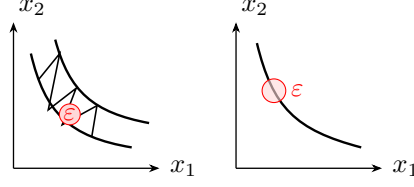


Figure 1: The first implication of local non-satiation, drawn as in the original. *Left:* a thick indifference zone; a small ε -ball inside it contains only bundles indifferent to its centre, so there is no strictly better bundle nearby — local non-satiation rules this out (“kill the thick zone”). *Right:* a thin indifference curve; every ε -ball around a point on it reaches bundles above the curve.

5 Why the budget constraint binds

By contradiction. Suppose the optimal bundle x does not exhaust the budget, $p \cdot x < w$. Then every bundle close enough to x is still affordable (there is an $\varepsilon > 0$ such that $p \cdot y < w$ whenever $\|y - x\| < \varepsilon$), and by local non-satiation one of them, y , has $y \succ x$. So something better was affordable. This contradicts the fact that the original bundle is optimal; hence $p \cdot x = w$.

6 Monotonicity and convexity

Axioms with more content regarding the structure of preferences.

- *Monotonicity:* $\succsim$ is monotone if $x \gg y$ implies $x \succ y$.
- *Strong monotonicity:* $\succsim$ is strongly monotone if $x \geq y$ and $x \neq y$, then $x \succ y$.

Strong monotonicity implies monotonicity, and (on $X = \mathbb{R}_+^L$) monotonicity implies local non-satiation, because the consumer would always like more of everything: $x + \frac{\varepsilon}{2\sqrt{L}}(1, \dots, 1)$ is within ε of x and strictly preferred to it.

- *Convexity:* $\succsim$ is convex if, for every x , $y \succsim x$ and $z \succsim x$ imply

$$\alpha y + (1 - \alpha)z \succsim x \quad \text{for all } \alpha \in [0, 1].$$

Avr. (average) is always at least as good as extreme. Equivalently, for every x the superior (upper contour) set is a convex set,

$$\{y \in \mathbb{R}_+^L : y \succsim x\}.$$

In a picture: indifference curves are bowed towards the origin (a diminishing marginal rate of substitution).

7 Strict convexity

Strict convexity. $\succsim$ is strictly convex if, for every x , $y \succsim x$, $z \succsim x$ and $y \neq z$ imply

$$\alpha y + (1 - \alpha)z \succ x \quad \text{for all } \alpha \in (0, 1).$$

Indifference curves then have no flat segments; with strictly convex (rational, continuous, locally non-satiated) preferences the demand $x(p, w)$ is single-valued, a demand function.

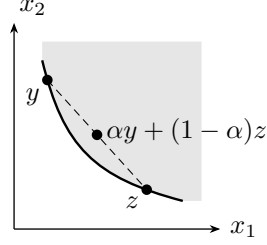


Figure 2: Convexity, drawn as in the original: y and z lie on the indifference curve through x , and their average $\alpha y + (1 - \alpha)z$ lies in the superior set of x (hatched in the notes, shaded here), which is convex.

8 Utility functions

The preference relation $\succsim$ is abstract; one may become used to thinking about utility functions. A utility function is a map $u : X \rightarrow \mathbb{R}$ with

$$x \succsim y \iff u(x) \geq u(y).$$

The utility function is just a made-up construct that is invented to represent preferences in a different way. Utility functions are often more convenient to work with, especially if we add assumptions such as differentiability.

Two facts follow directly from the definition. A utility function is *ordinal*: if u represents $\succsim$, so does $f \circ u$ for any strictly increasing $f : \mathbb{R} \rightarrow \mathbb{R}$, so only the ranking matters, not the numbers. And only a rational $\succsim$ can be represented: $\geq$ on $\mathbb{R}$ is complete and transitive, so the $\succsim$ it represents must be too.

9 When a utility representation fails

However, not all $\succsim$ can be represented by a utility function, even when they are rational. The most famous example is lexicographic preference: put the bundle with the most x_2 , and compare x_1 only to break ties,

$$x \succ y \iff x_2 > y_2, \text{ or } x_2 = y_2 \text{ and } x_1 > y_1.$$

No two distinct bundles are indifferent, and there is no utility function for it. It is not continuous: $x^n = (1, 1 + 1/n) \succ y = (2, 1)$ for every n , but in the limit $x = (1, 1) \prec y$.

To use a utility function, more assumptions are needed. At least, the extra assumptions have a natural economic interpretation. We need continuity.

10 Continuity

Continuity. $\succsim$ is continuous if for any sequence $\{(x^n, y^n)\}_{n=1}^\infty$ with $x^n \succsim y^n$ for all n , and $x = \lim_{n \rightarrow \infty} x^n$, $y = \lim_{n \rightarrow \infty} y^n$, it holds that $x \succsim y$.

That is: no sudden preference reversal. Equivalently, for every x the upper and lower contour sets $\{y : y \succsim x\}$ and $\{y : x \succsim y\}$ are closed. The payoff (Debreu): a rational and continuous $\succsim$ can be represented by a continuous utility function. Lexicographic preferences fail exactly this continuity.

Transcription notes

- Page 1 of the original is a title page reading “Lesson 6 : Chap. 3 — classical demand Th.”; the abbreviation is read as *theory*, and the page carries no other content. The subtitle previously read “Part 2”; it now follows the title page (Chapter 3).
- “this hand written into the consumer” has no verb in the original; it is read as *This is* hard-wired into the consumer (“hand written” is taken as a mishearing of “hard-wired”, the usual phrase).
- “captures evenly about his taste or preference” is read as *captures everything* about his taste or preference — the ending of the word is written as if it were “-ly”.
- “we start with axioms imply that the consumer’s pref. are relations in basic sense” is ungrammatical as written (the finite verb “imply” has no relative pronoun); it is read as *axioms that imply . . .* “relations in basic sense” is kept as written; in the notation of the course these are binary relations on X in the ordinary set-theoretic sense.
- “indifference curve are thin” is corrected to *curves*; “kill the thick zone” is written in red ink in the original.
- Local non-satiation is written with $y > x$ in the original; $y \succ x$ is used here, as elsewhere in these notes the strict relation is the hook symbol.
- **Corrected (earlier transcription):** page 4 draws two sketches under “Two important implications”: on the left a thick indifference zone (a band hatched with a zig-zag) with a small red ε -ball inside it, on the right a thin indifference curve with a red ε -ball on it. Both red marks are ε -balls (the ε is legible at high zoom), not a “circled slash”. The figure previously had the panels swapped, a red slash, and invented labels x, x' ; it now follows the original.
- “By contradiction, the budget is not exhausted” states the conclusion of a contradiction argument without spelling out the assumption; the sentence is read as *suppose* the budget is not exhausted. The original also omits the verb in “mean. that you can find something better”, read as *meaning* that.
- “rehanding” $\rightarrow$ *regarding*; “AVR.” is read as *average*.
- **Corrected:** the convexity line was written with three conditions before the implication ($y \succsim x, y \succsim z, z \succsim x$) and no range for α . The middle condition is redundant, and it is dropped; the definition is typeset in the standard form, $y \succsim x$ and $z \succsim x$ imply $\alpha y + (1 - \alpha)z \succsim x$ for all $\alpha \in [0, 1]$. “Average is always better than extreme” is softened to *at least as good as*: convexity is a weak relation.
- The superior set is written $\{y \in \cdot \mid y \succsim x\}$ in the original; the set after $\in$ is an unclear glyph (like “[”), read as the consumption set $\mathbb{R}_+^L$. The convexity sketch marks three dots on the indifference curve region (two on the curve, their average inside the hatched set), reproduced as y, z and $\alpha y + (1 - \alpha)z$; a small pair of arrows drawn near the origin of that sketch is illegible and is omitted.
- “stric convexity $\succsim$ is strict $\succsim$ is strictly convex” repeats the phrase; only *strictly convex* is kept.
- **Corrected:** strict convexity is written “if $y \succ x, y \succ z, z \succ x$ implies $\alpha y + (1 - \alpha)z \succ x \forall \alpha \in (0, 1)$ ”. As with convexity, the middle condition is redundant and is dropped; the condition $y \neq z$ is missing and is added (without it $y = z = x$ would give $x \succ x$).
- **Added:** a precise proof that the budget constraint binds, the chain strong monotonicity $\Rightarrow$ monotonicity $\Rightarrow$ local non-satiation, the ordinal nature of utility and why representability requires rationality, the formal lexicographic order with a sequence showing it is not continuous, Debreu’s representation result for continuous preferences, the equivalent closed-contour-set form of continuity, the remark that convex indifference curves are bowed toward the origin, and the remark after strict convexity (no flat segments; single-valued demand).
- “is abstract, to may become use to thinking about utility functions” is read as *one may become used to thinking about utility functions*; “can be represented at a utility func.” is read as *by* a utility function.
- Lexicographic preference is written as “put the bundle with the most x_2 ”. The order is normally described by comparing the first coordinate first; the notes’ order (good 2 first) is kept, and the formal definition is written in that order.

- In the continuity definition the original labels the sequence $\{(x^n, y^n)\}_{N=1}^{\infty}$ and then writes “ $\forall N$ ” before switching to lower-case n in the limits. The index is written n throughout here.
- “the extra assumptions has a nature econ interpretation” is read as *have a natural* economic interpretation.
- Page 9 of the original carries a single line, “No sudden preference reversal.”, which closes the continuity section.
- Pages of the original, for tracing back: page 2 the classical approach and the preference relation; page 3 completeness, transitivity, and local non-satiation; page 4 the two implications with the thick-zone sketch; page 5 the binding budget constraint and monotonicity; page 6 strong monotonicity, convexity and the superior set; page 7 strict convexity and the utility function; page 8 lexicographic preference and continuity; page 9 the closing line.